SOPAAN

Every step, signed.

Unified System Design Document

Offline-first learning-to-livelihood platform for rural India

High-level design, low-level design, architecture diagrams, principles

Sopaan (Hindi: stairway, steps). Each step a student completes becomes a signed, portable, verifiable record that works with or without internet. The platform merges three proposals (**Pagdandi**, **DISHA**, **Sanad**) into one design with one vocabulary and one trust chain.

Item	Detail
Event	Citi Ada Lovelace Hackathon: Reimagining Education and Economic Empowerment for Rural India
Version / date	1.0, 14 October 2026
Backend stack	Spring Boot (trust core, sync, ledger) + FastAPI (AI and tooling); everything on free tiers
Status	Proposed design. Every number marked "target" is a proposal to be measured, not a result.
Name check	One web search found no obvious existing product called Sopaan. Run a trademark, domain and app-store check before use.

Contents

0. Document guide and lineage . . . 4

0.1 What each proposal contributed . . . 4

0.2 How to read this document . . . 4

1. Executive summary . . . 5

1.1 The five-link chain (the product in one picture) . . . 5

1.2 Eight decisions that shape everything . . . 5

2. Requirements . . . 6

2.1 Traceability: the brief and the meeting insights . . . 6

2.2 Functional requirements . . . 6

2.3 Non-functional requirements (all are targets to be measured) . . . 6

2.4 Constraints and non-goals . . . 7

3. Design principles and the distributed-systems toolkit . . . 8

3.1 Ten principles . . . 8

3.2 System-design concepts and how Sopaan uses them . . . 8

3.3 Consistency model per record type (PACELC position) . . . 9

4. High-level design (HLD) . . . 10

4.1 System context . . . 10

4.2 Three tiers, each useful without the one above . . . 10

4.3 The trust chain . . . 11

4.4 Logical components and responsibilities . . . 11

4.5 Technology stack and roles . . . 12

4.6 Deployment on free tiers (verified 14 October 2026; re-check before submission) . . . 12

4.7 What works offline and what needs a connection . . . 13

5. Low-level design (LLD) . . . 14

5.1 Student PWA . . . 14

5.2 Spring Boot backend . . . 15

5.3 FastAPI AI and tooling service . . . 17

5.4 Wire formats . . . 18

5.5 Algorithms . . . 18

5.6 Key sequences and state machines . . . 19

5.7 Hub profile and transports . . . 21

6. Failure, security and responsible-AI analysis . . . 22

6.1 Failure modes and responses . . . 22

6.2 Threat model (summary) . . . 22

6.3 Privacy, child safety and fairness . . . 22

7. Scalability and capacity . . . 24

7.1 Capacity estimate for a first pilot . . . 24

7.2 Scaling triggers and responses . . . 24

7.3 Evolution path . . . 24

8. Quality: testing, metrics and observability . . . 25

8.1 Test strategy . . . 25

8.2 Metrics and acceptance criteria (targets, not results) . . . 25

8.3 Observability . . . 25

9. Delivery: scope, plan, demo and risks . . . 26

9.1 Scope tiers . . . 26

9.2 Build order . . . 26

9.3 Live demo: Radha's story . . . 26

9.4 Risk register . . . 26

10. Business model, Citi alignment and judging map . . . 27

10.1 Revenue model (illustrative planning assumptions to validate in a pilot) . . . 27

10.2 Mapping to the judging criteria . . . 27

Appendices . . . 28

A. Glossary . . . 28

B. Pre-submission checklist . . . 28

C. Verification notes . . . 28

0. Document guide and lineage

This document is the single source of truth. It replaces the three earlier drafts. Where they disagreed, the decision and the reason are recorded below, so nothing in the design is there by accident.

0.1 What each proposal contributed

Source	Kept in Sopaan	Changed or cut
Pagdandi	Radha narrative; evidence chain (rule, result, step, credential, credit); Next-Step Card; Family Voice; outcome-linked credits; five bounded agents; failure-mode table; business model; metrics; pre-submission checklist.	Next.js replaced by Vite + Preact (lighter offline PWA). LAN hub sync demoted to a stretch goal.
DISHA	Discover / Learn / Earn / Prove framing; consistency strategy per record type; three-outcome eligibility; explicit MVP exclusions; tamper and stale-registry demo cases; "targets, not results" discipline.	Name and document structure merged away; nothing unique left.
Sanad	Trust lives in the signature, not the channel; signed packs over any channel; tiered voice and inference; facilitator hub as real infrastructure; credits bounded by cap and counter; append-only logs; Bloom filter for revocations.	Cut: Rust/WASM core, CRDT library, WebRTC mesh, Merkle anchoring, Rego, Monte Carlo twin, on-device LLM, and the claim that credentials act as a credit signal.
System design notes (provided)	Used as the design vocabulary: WAL, segmented log, high watermark, lease, fencing, Bloom filter, checksum, version vectors, hinted handoff, read repair, leader-follower, CAP/PACELC. See section 3.2.	Not used where they would be decoration: consistent hashing, gossip, phi-accrual, multi-node quorum.

0.2 How to read this document

- **Sections 1 to 3**: what and why (summary, requirements, principles).
- **Section 4**: high-level design (context, tiers, trust chain, stack, deployment).
- **Section 5**: low-level design (client, server, FastAPI, data model, wire formats, algorithms, sequences).
- **Sections 6 to 9**: failure and security analysis, scale, quality, delivery plan and risks.
- **Section 10 and appendices**: business model, judging map, glossary, checklist, verification notes.

1. Executive summary

Radha is in Class 10 in rural Maharashtra. She is good with numbers and sewing machines, has never heard of a polytechnic diploma or an apprenticeship scheme, and her teacher supports 60 students on a patchy 3G connection. The information she needs exists, but it cannot be trusted, proven or funded as one journey. Three things break:

- **Guidance cannot be trusted.** Scheme rules are scattered and change often, and a chatbot can sound sure and be wrong.
- **Skills cannot be proven.** Informal skills have no portable, verifiable evidence.
- **Funding cannot follow results.** Sponsors fund activity because they cannot verify completion.

Sopaan closes the loop with one chain of checkable records:

Core thesis. AI understands the student, reviewed rules decide eligibility, digital signatures prove skills, sponsor credits release only against verified progress, and the edge keeps working when the network does not.

1.1 The five-link chain (the product in one picture)

Link	Name	What it is
1	Signed rule pack	Reviewed, versioned scheme rules with source, date and reviewer signatures
2	Eligibility result	Computed on device; ELIGIBLE, NOT ELIGIBLE (rule) or NEED INFO; cites the exact rule version
3	Next-Step Card	Documents, deadline, place and helper, plus optional Family Voice audio
4	Signed credential	Skill Passport entry signed by an authorised, leased issuer; verifiable offline by QR
5	Credit release	Sponsor credit unlocked only by a valid credential; capped, counted, reconciled at sync

1.2 Eight decisions that shape everything

#	Decision	Why
D1	Trust lives in the signature, not the channel.	Packs, issuer lists, credentials and credit tokens are Ed25519-signed artifacts. They can travel by cloud sync, QR, WhatsApp file, Bluetooth or pendrive and are verified identically. Fits rural reality and needs no special infrastructure.
D2	AI proposes, rules decide.	A wrong "you qualify" costs a family a trip and a deadline. Eligibility is deterministic, versioned and cited; the LLM only extracts, explains and drafts rule proposals that a human must approve.
D3	Local database is the system of record for a session.	The device never waits for the server. The cloud is a reconciler, not a gatekeeper. Render cold starts become invisible.
D4	Modular monolith (Spring Boot) plus one optional AI service (FastAPI).	Fewest moving parts that satisfy the requirements. FastAPI can be switched off without losing any decision, credential or credit.
D5	One backend, two profiles: "cloud" and "hub".	The facilitator laptop runs the same jar and the same tests as the cloud. No second implementation to maintain.
D6	Append-only signed event log plus idempotency, not a CRDT library.	A grow-only event set already merges by union. Per-field version vectors cover the few mutable fields. Much less to build and prove.
D7	Free tier first.	Static host, Render free, Neon free, GitHub Actions, open-source models. Limits and mitigations are in section 4.6.
D8	Every result carries its version and its age.	Rule pack version, registry sync time and pack expiry are always visible. Offline availability is never presented as global truth.

2. Requirements

2.1 Traceability: the brief and the meeting insights

Requirement (source)	Design response	Where
Voice in/out in local language, simple UI for low literacy (brief)	Closed-grammar offline STT, icon confirmation, pre-recorded + device TTS, text fallback	5.1.4
Low-end Android, offline, low-bandwidth sync (brief)	PWA, IndexedDB, WAL outbox, small packs, batch sync, signed offline channels	4.7, 5.5
Facilitator mode: one teacher, many students (brief)	Hub profile: cohort dashboard, issuing, queued events, module authoring	4.2, 5.7
AI grounded in real data (brief)	Reviewed scheme and catalogue packs with sources; LLM never invents rules	5.3
Meaningful LLM use: profiling, matching, eligibility, pathways (brief)	LLM where language is uncertain (profile extraction, rule extraction, explanation); deterministic code where correctness matters	3.1, 5.3
Open-source, extensible, standards-based APIs (brief)	Open stack, versioned packs, OpenAPI, signed-envelope format, adapters for partners	4.5, 5.2
Monetization (brief, 15%)	Students free; NGO/CSR, employer, provider, facilitator revenue	10
Scalability as evaluation (meeting)	On-device heavy work, event-driven server, measured scaling triggers	7
Secure; responsible and ethical AI (meeting)	Threat model, key management, fairness tests, human review	6
Simplicity; AI with real purpose; metrics in submission (meeting)	One chain, five links; each AI use justified; metrics table	1, 8

2.2 Functional requirements

ID	Requirement
FR1	Voice or icon profile: student answers, system repeats back with icons, student confirms; every field carries confidence and source.
FR2	Career discovery over a broad catalogue (vocational, technical, academic, enterprise); at least three clusters shown when they exist.
FR3	Pathway plan: ordered steps (qualification, training, documents, deadlines) built from a prerequisite graph; progress only from recorded events.
FR4	Eligibility for each scheme returns ELIGIBLE, NOT ELIGIBLE (named rule) or NEED INFO (fields/documents), with source, rule version and date.
FR5	Next-Step Card per opportunity plus optional Family Voice audio summary in the family language.
FR6	Skill Passport: authorised assessor signs a credential; anyone verifies by QR offline with a visible registry age.
FR7	Facilitator dashboard: roster, missing documents, assessments, credential issue, module authoring, revenue share.
FR8	Outcome-linked credits: sponsor pool, signed allowance lease, offline redemption within cap, reconciliation at sync.
FR9	Signed pack update over any channel (cloud, file, QR); atomic swap; automatic rollback on bad signature.
FR10	Reviewer workflow: LLM-proposed rules are inert until two reviewers co-sign and golden tests pass.
FR11	Consent capture and guardian consent for minors; deletion on request; audit trail for consequential actions.
FR12	Sponsor/CSR aggregate dashboard with no identifiable student data.

2.3 Non-functional requirements (all are targets to be measured)

Quality	Target	How it will be measured
Offline completeness	100% of "Must" flows complete in real airplane mode after install	Playwright offline context + real device test
Initial load	App shell + core pack <= 2 MB gzipped; voice model is a separate optional download	Bundle report in CI
Cold start (2 GB Android)	< 5 s to interactive	Lighthouse + real low-end device
Credential size	<= 1,000 bytes in a single QR	Unit test on encoder
Offline verify time	< 200 ms on a mid-range phone	Benchmark in test harness
Sync correctness	No duplicate business effect for a repeated event_id; >= 99% of valid queued events acked in fault tests	Idempotency + fault-injection tests
API latency (warm)	p95 < 500 ms; cold start (~1 min on Render free) is excluded and hidden by offline-first design	k6 load test
Security	All defined unauthorised-access test cases denied; tampered artifacts rejected	Authorisation + crypto tests, ZAP baseline

Quality	Target	How it will be measured
Privacy	No raw audio leaves the device by default; pseudonymous student IDs; no Aadhaar by default	Data-flow review, network tests
Freshness	Warning shown whenever a pack or registry is past its expiry	UI tests with clock control
Cost	Variable cloud and AI cost < Rs 3 per student per month (planning target)	Provider quotes + usage logs

2.4 Constraints and non-goals

- **Constraints:** everything on free tiers; Spring Boot + FastAPI backend; app must work offline with minimal internet dependence; open-source components.
- **Non-goals for the prototype:** live bank settlement or UPI payouts, official award decisions, an LLM running on low-end phones, credentials as credit scores, any partner integration that is not confirmed. These appear only as clearly labelled "planned".

3. Design principles and the distributed-systems toolkit

3.1 Ten principles

#	Principle	Applied as
P1	Separate understanding from deciding.	LLM and speech sit outside the trust core; rules engine and signature checks are deterministic code with golden tests.
P2	Data as code, signed and versioned.	Rules, catalogue, registry and revocations ship as signed packs; a scheme change is a reviewed data push, not an app release.
P3	Offline-first, sync second.	All Must flows read and write the local store; the network is an optimisation.
P4	Make every write idempotent.	UUIDv7 event IDs, unique constraint server-side, ack by high-water mark; retries are always safe.
P5	Honest consistency.	Per-record-type model (table 3.3). Disconnected devices never claim global agreement; staleness is shown.
P6	Bound the blast radius.	Credits capped per device and lease; issuer keys leased with expiry and epoch; packs expire.
P7	Fail soft, degrade visibly.	AI down: templates. Registry old: labelled. Pack bad: keep previous. Never a blank screen, never a silent guess.
P8	Human in the loop where harm is possible.	Rule approval (2-of-3 reviewers), profile confirmation by the student, facilitator can annotate but not override a rule silently.
P9	Minimise data; pseudonymise.	Random student IDs, roster held by the facilitator, consent versioning, aggregate-only sponsor view.
P10	Simplest thing that meets a measured need.	Modular monolith, one DB, no Kafka/Redis/vector DB until a measurement demands it.

3.2 System-design concepts and how Sopaan uses them

The provided system-design note lists the standard toolkit. The table states, for each concept, whether Sopaan uses it, where, and why. Concepts that do not fit are marked "not used" with the reason; naming them without using them would be decoration.

Concept	Use	Where and why
Scalability	Yes	Heavy work runs on devices; server load grows with sync events, not taps. Triggers and path in section 7.
Availability / reliability	Yes	Local-first reads and writes; graceful degradation table in 6.1.
Redundancy	Partial	Packs and registry replicated on every device and hub; cloud is single-region on free tier (honest gap, planned multi-AZ).
Load balancing	Planned	Not in free-tier prototype; stateless API workers behind a managed balancer in the pilot.
Fault tolerance	Yes	Atomic pack swap with rollback, durable outbox, hub forwards later, AI service optional.
Caching	Yes	Service worker caches shell and packs; hub caches packs for the cohort; ETag on pack endpoints.
Sharding / consistent hashing	Planned	Future shard key = district_id or student_id hash for ingest. Not needed at pilot scale.
Microservices / SOA	Light	Two services only (Spring, FastAPI) with an explicit REST contract; no further splitting without a measured need.
Bloom filter	Yes	Compact revocation set on devices: a negative is definitive, a positive means "possibly revoked, check exact list or go online" (5.5.5).
Quorum	Governance only	2-of-3 reviewer co-signatures on rule packs. There is no multi-replica quorum read/write path.
Leader and follower	Yes	Cloud is leader for consent, ledger and registry; hubs and devices are followers for packs and read-mostly state.
Write-ahead log	Yes	Client outbox is a WAL (write event and state change in one transaction); Postgres WAL on the server.
Segmented log	Yes	Outbox partitioned into daily segments; segments below the high-water mark are dropped as a unit.
High watermark	Yes	Per-device contiguous acked sequence number returned by the server; the client prunes up to it.
Lease	Yes	Issuer keys and credit allowances are time-limited leases that must be renewed at sync; expiry is enforced offline.
Heartbeat	Light	Hub and device "last seen" stamped on each sync; dashboard shows stale hubs. No peer cluster to monitor.
Gossip, phi-accrual	Not used	There is no peer cluster to disseminate to or to detect failure in; hub-to-cloud is a simple client-server link.

Concept	Use	Where and why
Split brain	Mitigated	Two hubs must not issue under one key: one issuer lease per key, per-issuer counters, epoch check at verify.
Fencing	Yes	Key epoch acts as a fencing token: credentials signed under an older epoch are rejected once the registry epoch advances.
Checksum	Yes	SHA-256 evidence hashes and pack manifests; QR has built-in error correction; signature covers everything that matters.
Vector clocks	Yes (light)	Per-field version vectors {device: counter} for the few mutable profile fields; concurrent edits are flagged, not silently overwritten.
Hinted handoff	Yes (analogue)	Hub holds events for the cloud while the cloud is unreachable and forwards them later.
Read repair / anti-entropy	Yes (analogue)	Pull step reconciles diverged state (pack versions, registry, profile conflicts) when the client next reads.
Merkle tree	Partial	Pack manifest is a hash list under one signature (Merkle-lite) enabling delta updates; a full tree only pays off with thousands of files. No blockchain.
CAP / PACELC	Yes	Choose availability for cached guidance and verification during partitions; choose consistency for consent, ledger and revocation at the cloud. Table 3.3.

3.3 Consistency model per record type (PACELC position)

Record	During partition (P)	Normal operation (E)	Mechanism
Rule pack, catalogue	Available (cached, signed)	Latency over consistency (pull on sync)	Signed versioned snapshot, expiry, rollback
Issuer registry, revocations	Available but labelled stale	Consistent at cloud (leader)	Signed snapshot + epoch + Bloom filter
Credentials	Available (immutable)	Immutable, no conflict	Signature + issuer + expiry + epoch
Learning progress, assessments	Available	Eventual	Append-only events, idempotent by event_id
Profile fields	Available	Eventual with conflict flag	Per-field version vector; student confirms conflicts
Consent	Cloud-authoritative (queue request)	Consistent	Server-side write, never merged blindly
Credit redemption	Available within lease cap	Consistent at settlement	Cap + counter + dedup; violations flagged

4. High-level design (HLD)

4.1 System context

Five human roles interact with Sopaan. Reviewers are a governance role that keeps AI-proposed rules honest. External providers are optional and only reachable online; integrations with government and payment systems are explicitly planned, not claimed.

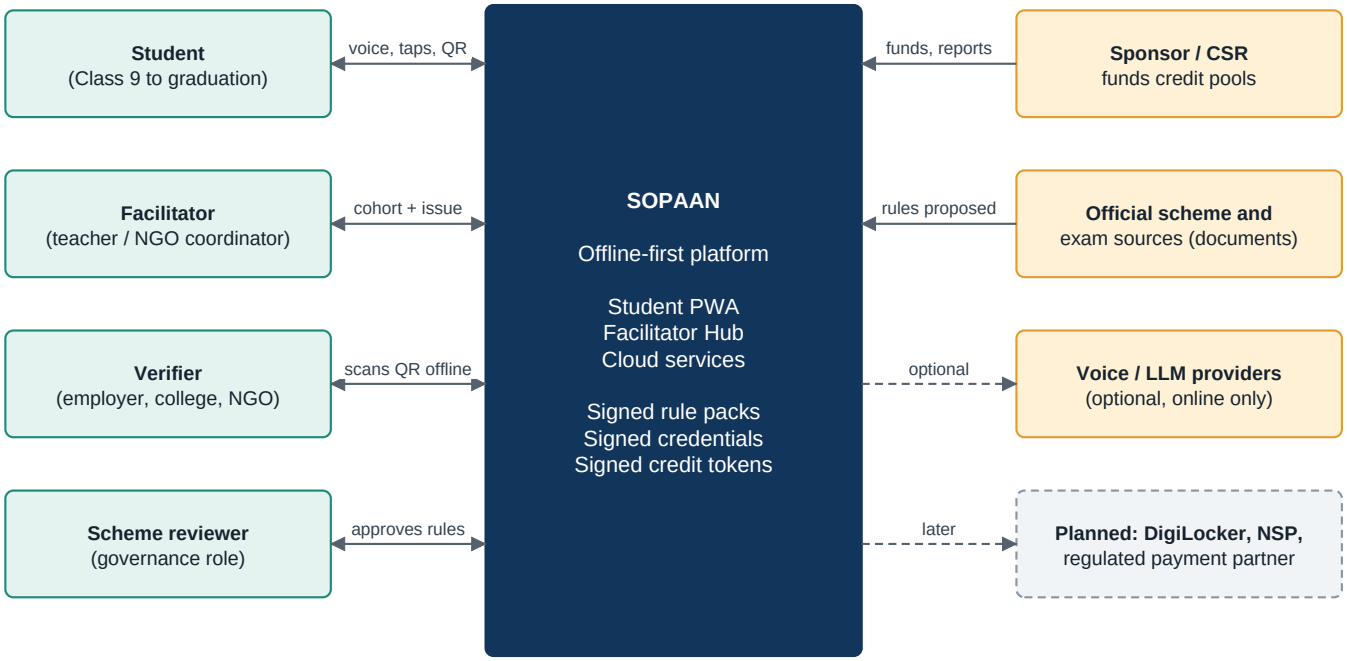


Figure 1. System context. Dashed boxes are optional or planned and are not required for any Must flow.

4.2 Three tiers, each useful without the one above

The **student device** works alone after install. The **facilitator hub** is the same backend running on a laptop, adding cohort management and issuing. The **cloud** is the system of record and the only place that reconciles everyone. Information moves up as signed events and down as signed packs, over any channel.

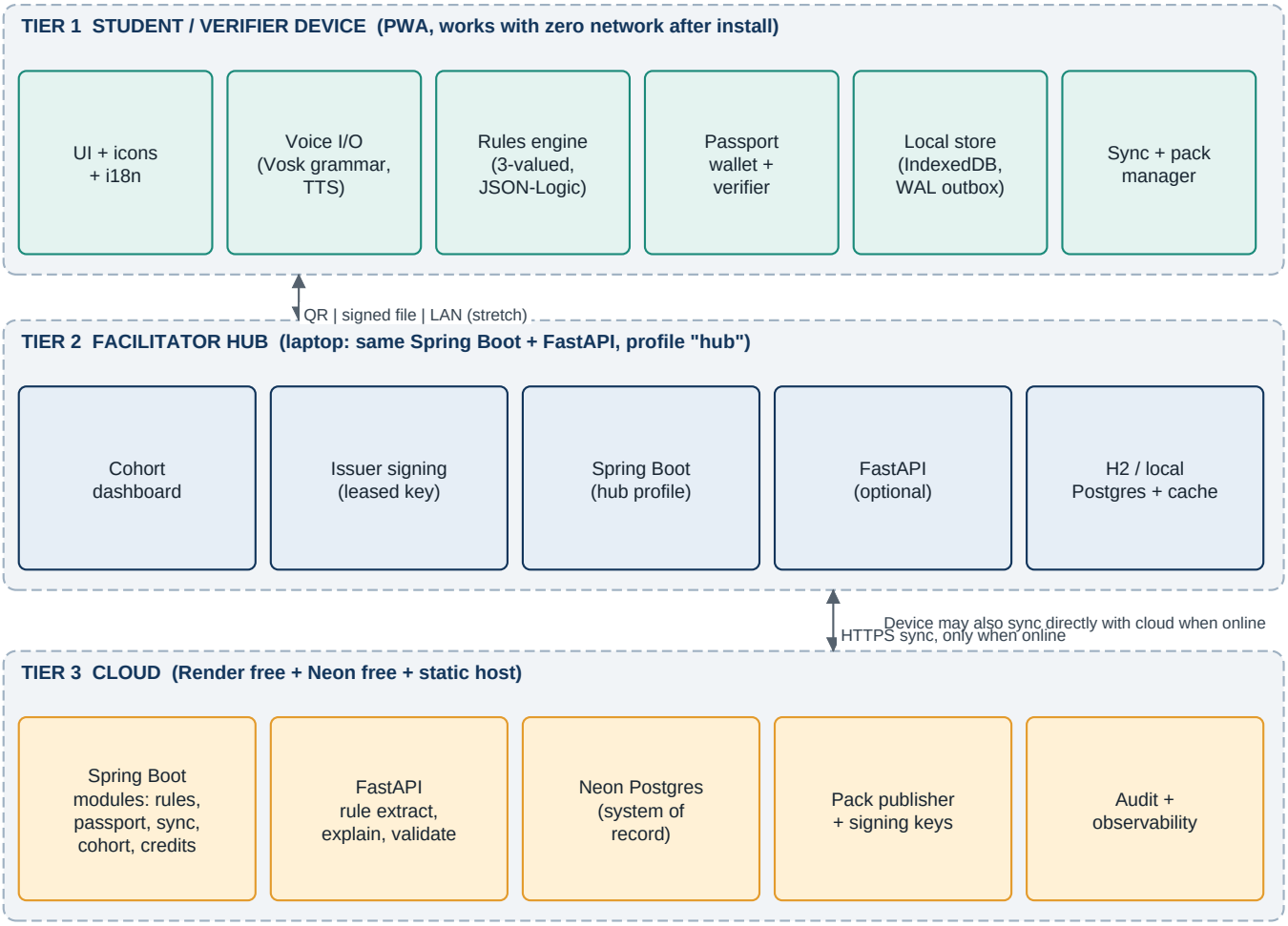


Figure 2. Tiered architecture. Arrows show the transport options; every transport carries signed artifacts.

4.3 The trust chain

Each link produces a record the next link can check. That is how Sopaan can say something a chatbot cannot: this answer came from this reviewed rule, and this result was proven by this issuer.

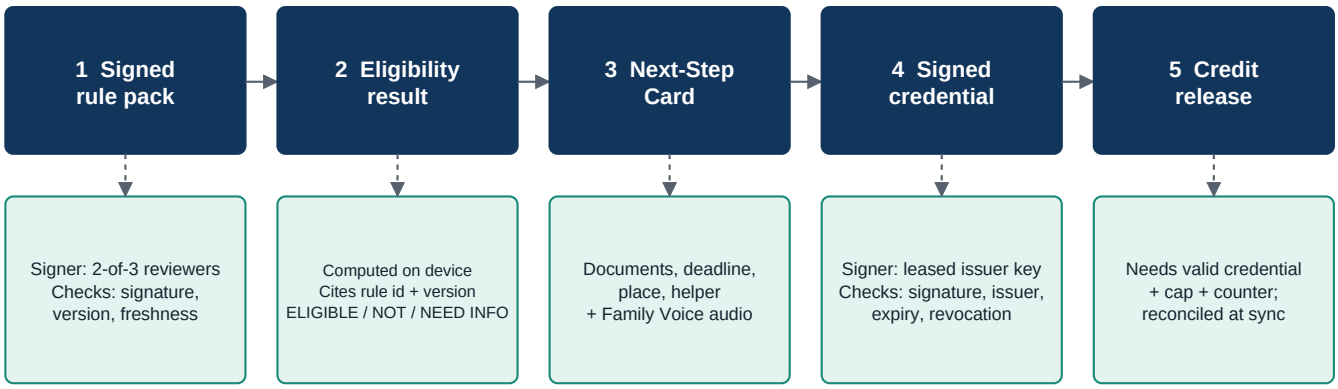


Figure 3. The five-link trust chain with signer and verification at each link.

4.4 Logical components and responsibilities

Component	Tier	Responsibility	Never does
Student PWA	Device	Profile, discovery, pathway, eligibility, wallet, verify, credits, sync	Trust unsigned packs; decide with an LLM
Hub (Spring + FastAPI, profile "hub")	Laptop	Cohort dashboard, issue credentials under a leased key, cache packs, forward events	Issue without a valid lease; edit rules

Component	Tier	Responsibility	Never does
rules module	Spring	Pack lifecycle: propose, review, golden-test, sign, publish, roll back	Ship a rule an LLM wrote without human approval
passport module	Spring	Issuer certs, credential issue, revocation, signed registry snapshot, Bloom builder	Treat a valid signature as proof the assessment was right
sync module	Spring	Idempotent event ingest, high-water marks, conflict flags, pull manifests	Merge consent blindly
cohort module	Spring	Rosters, assessments, modules, facilitator revenue share	Expose identifiable data to sponsors
credits module	Spring	Pools, allowance leases, token settlement, double-spend detection	Move real money in the prototype
audit module	Spring	Append-only hash-chained log of consequential actions	Delete or rewrite entries
FastAPI service	Cloud/hub	Rule extraction, profile extraction, explanation, pack validation, fairness harness	Write eligibility results or issue credentials

4.5 Technology stack and roles

Layer	Choice	Role	Why this one
PWA app	Vite + Preact + TypeScript + Workbox	UI, offline shell, service worker	Small bundle for 2 GB phones; first-class offline caching
Local data	IndexedDB via Dexie	Profile, packs, wallet, outbox	Available on every target browser; transactions give the WAL property
Rules (client)	json-logic-js + 3-valued wrapper	Eligibility on device	Declarative rules as data, same format as server
Rules (server)	json-logic-java + same wrapper	Eligibility on server, pack tests	Golden suite proves parity with the client
Crypto	@noble/ed25519, SHA-256; JDK Ed25519; PyNaCl	Signatures and hashes in all three runtimes	Pure JS works on old Android Chrome; no framework lock-in
Voice in	Vosk WASM with restricted grammar (verify Hindi model size and grammar support)	Offline slot recognition	Closed vocabulary beats dictation on small models; icons are the fallback
Voice out	Pre-recorded clips in packs + device speechSynthesis	Family Voice, read-back	Zero runtime cost; works offline
Backend core	Spring Boot 3, Java 17+	Trust core, sync, ledger, hub	Strong typing, mature security, team choice
AI service	FastAPI, Pydantic, optional LLM adapter	Rule extraction, extraction, explanation	Python AI ecosystem; isolated and optional
Database	Neon Postgres (cloud); H2 or local Postgres (hub)	System of record	Free, standard SQL, same migrations
Schema	Flyway	Versioned migrations	Same schema everywhere
Delivery	Docker Compose; GitHub Actions	Hub packaging; CI, pack signing, deploy	Free for public repos; reproducible demo
Testing	JUnit, fast-check, Playwright, k6, ZAP baseline	Unit, property, offline e2e, load, security	All open source

4.6 Deployment on free tiers (verified 14 October 2026; re-check before submission)

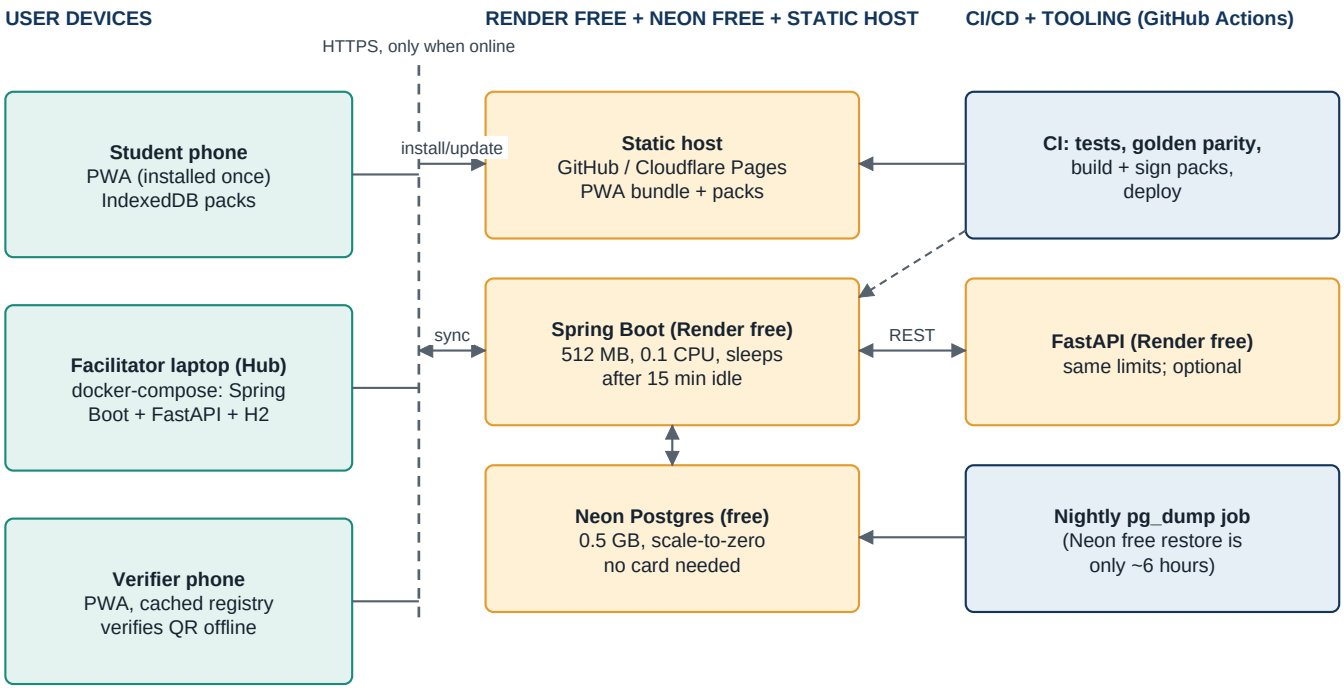


Figure 4. Deployment view. The same two services run on Render (cloud) and on the facilitator laptop (hub).

Need	Pick	Facts and consequences
Spring Boot host	Render free web service	512 MB RAM, 0.1 CPU, ephemeral filesystem; spins down after 15 idle minutes and takes about a minute to wake; 750 free instance hours per workspace per month. Mitigation: offline-first design hides cold starts; lean JVM (-Xmx256m, minimal starters); demo day runs locally.
FastAPI host	Second Render free service	Same limits. Hugging Face Docker Spaces are not relied on: current docs say compute Spaces need a paid plan.
Database	Neon Postgres free	0.5 GB storage and 100 compute-hours per project per month; no card; scale-to-zero. Avoid Render free Postgres (expires after 30 days). Neon free restore window is short, so run a nightly pg_dump.
Bigger VM (optional)	Oracle Always Free	The Arm allowance was halved in 2026 to 2 OCPU / 12 GB and capacity is often unavailable. A bonus, not the plan.
Static host	GitHub Pages or Cloudflare Pages	Serves the PWA and packs. Free; limits not re-verified here.
Cloud voice (optional)	Sarvam	Speech to text about Rs 30 per hour of audio; new-user credit documented as Rs 100 on current docs (other pages say Rs 1,000, so confirm). Used only to pre-generate audio clips at build time and as an online upgrade path; check commercial licensing for shipping generated audio.
CI	GitHub Actions	Free for public repositories; verify current minutes.

4.7 What works offline and what needs a connection

Works with zero network after first install	Needs a connection at some point
Voice or icon profile, career and pathway view	Fresh rule packs, catalogue and issuer registry (or a signed file from anyone)
Eligibility from the cached pack with source and version	New revocations beyond the cached snapshot
Next-Step Card and pre-recorded Family Voice audio	Event sync to the cloud and credit settlement
Credential issue (hub) and verification (any phone) by QR	LLM conversation, cloud speech, official submissions
Credit redemption within the allowance lease cap	Consent changes (cloud-authoritative)

Honest limits. The first install needs one online visit (or a sideloaded APK/TWA), because service workers need a secure context. A plain-HTTP page on a LAN IP cannot register a service worker, and an HTTPS page cannot call an HTTP LAN hub (mixed content). Phone-to-hub LAN sync is therefore a stretch goal that needs a locally trusted HTTPS setup; the prototype moves signed artifacts by cloud, QR and file.

5. Low-level design (LLD)

5.1 Student PWA

5.1.1 Components and module layout

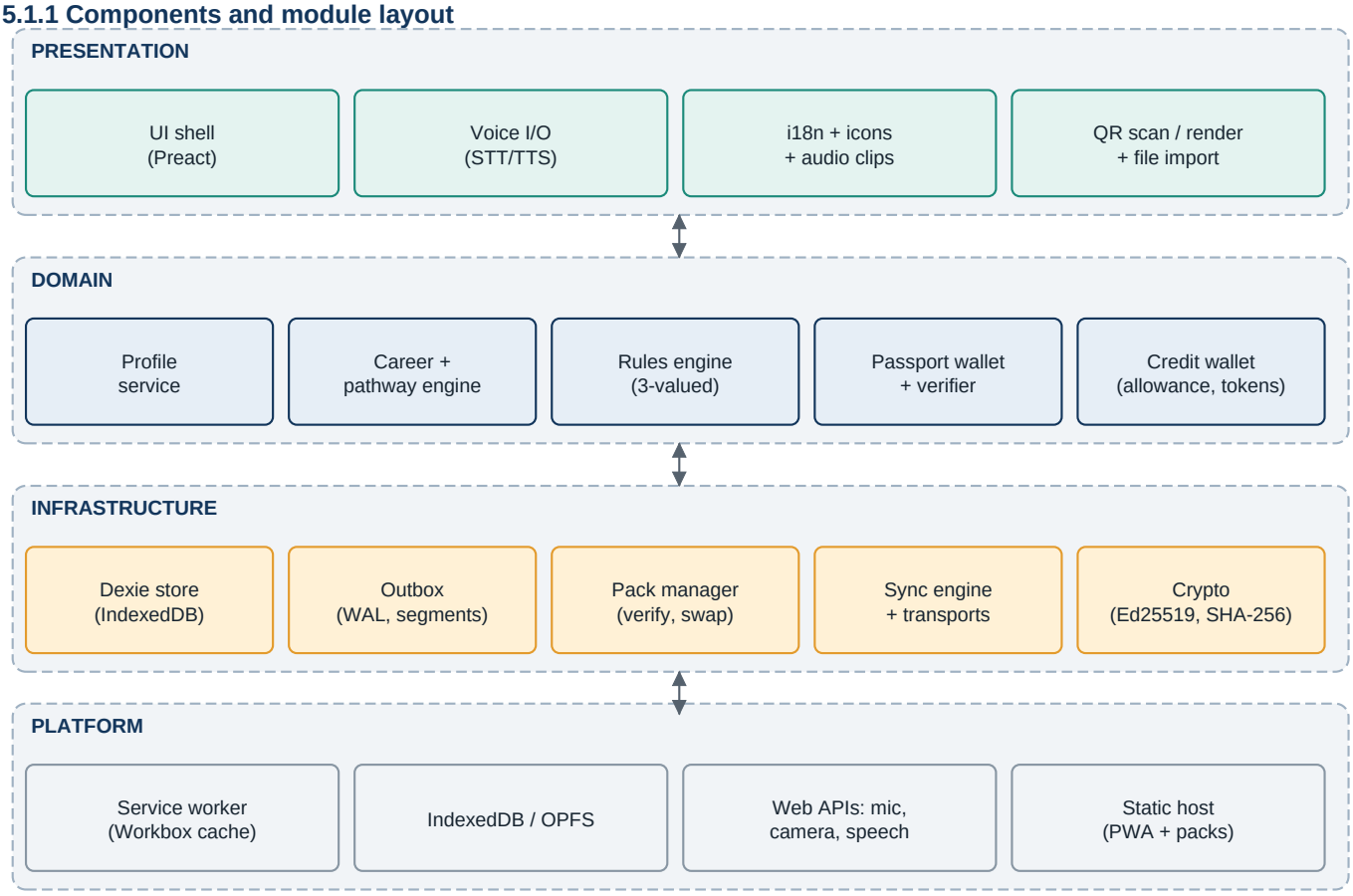


Figure 5. Client component diagram. Domain code depends only on the infrastructure layer interfaces.

```
src/
  core/        crypto.ts  envelope.ts  rules3v.ts  verify.ts  bloom.ts  ids.ts      (pure, no I/O, unit-tested)
  data/        db.ts (Dexie)  outbox.ts  packs.ts  profile.ts  wallet.ts  credits.ts
  features/    discover/  plan/  act/  prove/  credits/  hub/                        (screens + view-models)
  voice/       stt-vosk.ts  grammar.ts  normalize.ts  tts.ts  clips.ts
  sync/        engine.ts  transport-cloud.ts  transport-file.ts  transport-qr.ts
  sw/          service-worker.ts (Workbox: precache shell, runtime cache packs/audio)
  i18n/        hi.json  mr.json  icons.json
```

5.1.2 Local database (Dexie / IndexedDB)

```
db.version(1).stores({
  profile:      '&field, ver, updatedAt',           // value + version vector {deviceId: counter}
  events:       '&eventId, seq, segment, status',   // outbox WAL; status: QUEUED|SENT|ACKED|REJECTED
  packs:        '&name, version, expiresAt',        // rules, catalogue, audio index (signed envelopes)
  registry:     '&kid, epoch, validTo',             // issuer certs + bloom + syncedAt
  credentials:  '&credId, skill, issuedAt, expiry', // wallet
  allowance:    '&allowId, poolId, expiresAt',      // signed lease: cap, ctr_start
  tokens:       '&tokenId, allowId, ctr, status',   // credit tokens held / redeemed
  progress:     '&id, pathwayId, step',             // derived from events only
  keys:         '&kid',                             // device keypair (wrapped with PIN-derived key)
  settings:     '&k'
});
```

Every user action writes its state change and its event in **one IndexedDB transaction** (the WAL property). The outbox is split into daily **segments**; when the server high-water mark passes a segment, the whole segment is dropped. The device key is wrapped with a PIN-derived key (PBKDF2-HMAC-SHA256, at least 210,000 iterations, or scrypt via a vetted library). The PIN is access control, not a substitute for server-side authorisation.

5.1.3 Rules engine and career scoring

Eligibility. Each scheme is a tree of criteria whose leaves are JSON-Logic predicates. The wrapper evaluates every leaf to TRUE, FALSE or UNKNOWN (a required field is missing) and combines them with three-valued AND/OR (algorithm in 5.5.1). The result is one of three outcomes and always carries `rule_id`, `pack_version`, source URL and last-verified date.

Career discovery. A pure function over the catalogue pack, no LLM:

```
score(o) = 0.30*interest_match + 0.25*skill_overlap + 0.15*cost_fit + 0.15*distance_fit
         + 0.10*prereq_closeness - 0.05*gap_penalty
rerank   : greedy MMR over clusters {vocational, technical, academic, enterprise, services}
           pick top item, then next item maximising  score - 0.35*similarity_to_selected
guarantee: if >= 3 clusters have eligible options, show >= 3 clusters in the top 6
```

Pathway. A prerequisite DAG per occupation (qualification, training, documents, deadlines). A step is complete only when a matching `ProgressRecorded` event exists, never because any model said so.

5.1.4 Voice pipeline

```
mic -> AudioWorklet (16 kHz mono) -> Vosk WASM recognizer with slot grammar
    -> normalize (Hindi number words, category names, income bands, class)
    -> slot filler -> confidence = f(recognizer score, grammar match, repeat agreement)
    -> low confidence: re-prompt once, then icon picker (always available)
    -> read-back screen: icons + TTS of the interpreted value -> student taps confirm
    -> event ProfileFieldConfirmed {field, value, conf, source: "stt" | "icon" | "facilitator"}
```

- Closed-vocabulary grammars per question (class, category, income band, interests, skills) keep a small offline model accurate; whether a given Vosk build supports runtime grammars must be tested on the chosen model.
- Online upgrade: with consent, audio goes to a provider adapter and transcript to FastAPI `/profile:extract`; the same confirmation step follows. Raw audio never leaves the device by default.
- Output: pre-recorded clips (built once, shipped in the pack) for scheme explanations; device speechSynthesis for dynamic read-back; text always shown.

5.1.5 Sync and freshness states

The UI shows one badge per result: **Fresh** (pack within expiry, synced), **Cached** (within expiry, pending sync), **Stale** (past expiry; result still shown with warning), **Verified offline (registry as of date)** for credentials. These map one-to-one to the state machine in Figure 13.

5.2 Spring Boot backend

5.2.1 Components and packages

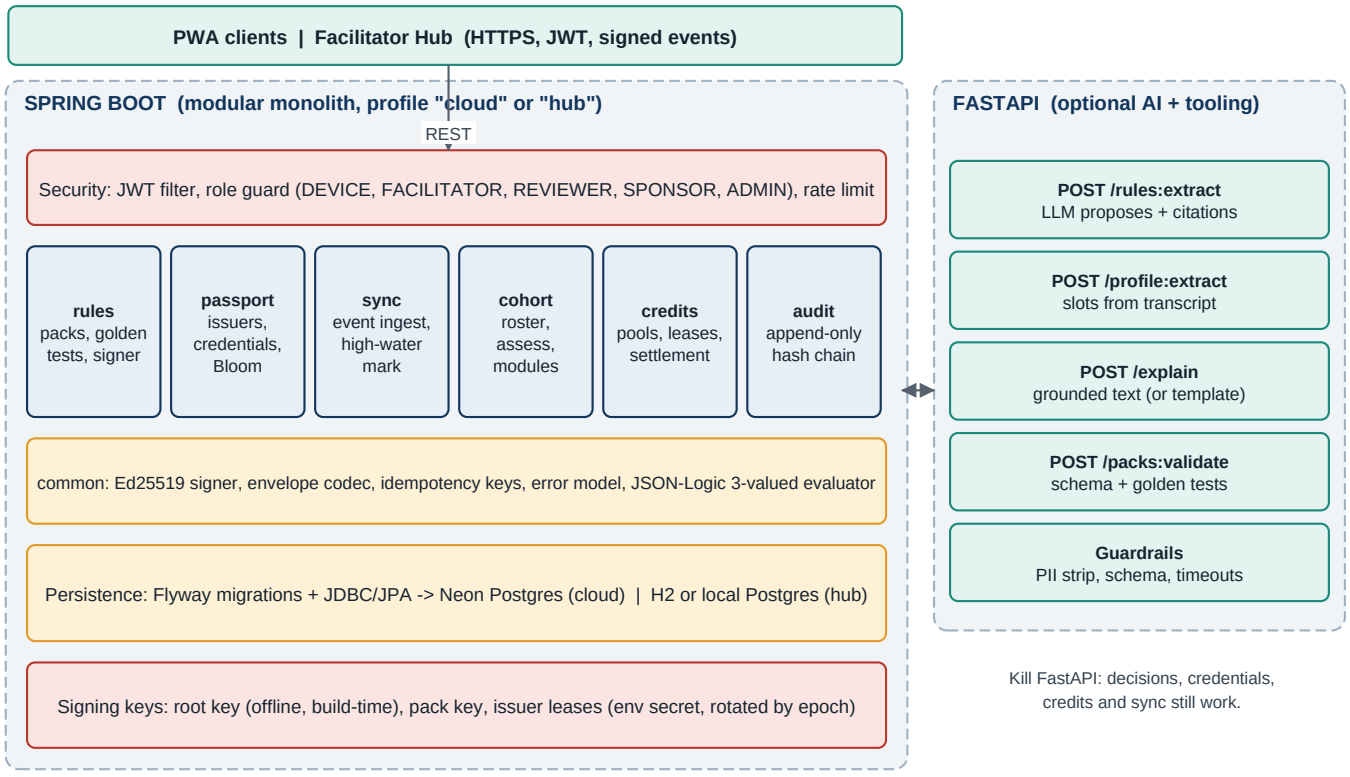


Figure 6. Cloud (and hub) components. The AI service is a replaceable side-car.

```
com.sopaan
  common/    Ed25519Signer  Envelope(Codec)  Ids(UuidV7)  JsonLogic3v  ErrorModel  IdempotencyGuard
  security/  JwtFilter  RoleGuard  DeviceAuth(sig challenge)  RateLimiter
  rules/     PackController  PackService  PackSigner  GoldenTestRunner  ReviewService(2-of-3)
  passport/  IssuerRegistry  CredentialService  RevocationService  BloomBuilder  RegistrySnapshotService
  sync/      SyncController  EventIngestService  HighWatermarkTracker  ConflictService  PullService
  cohort/    CohortService  AssessmentService  ModuleService  RevenueShareService
  credits/   PoolService  AllowanceService  SettlementService  DoubleSpendDetector
  audit/     AuditAppender (hash-chained)
  config/    Profiles: cloud | hub  (hub: H2, outbound sync target, no public listener by default)
```

Boundaries are enforced by package-private access and ArchUnit tests, so the monolith stays modular and a module can be extracted later if a measurement justifies it.

5.2.2 REST API (versioned, OpenAPI-described)

Endpoint	Role	Purpose and notes
POST /v1/devices:enroll	DEVICE	Register device public key; returns short-lived JWT. Pseudonymous student ID, no Aadhaar.
POST /v1/sync/push	DEVICE, HUB	<= 200 events per call; each event signed by device key; returns {acked_upto, rejected[], server_time}. Idempotent.
GET /v1/sync/pull?cursor=	DEVICE, HUB	Returns pack manifests, registry delta, conflicts to confirm; ETag supported.
GET /v1/packs/{name}/{version}	PUBLIC	Signed pack envelope. Anyone can mirror it; trust is the signature.
GET /v1/registry/snapshot	PUBLIC	Signed issuer certs, epoch, Bloom filter, exact revoked list.
POST /v1/credentials	FACILITATOR	Issue under a leased issuer cert; server countersigns into the registry log.
POST /v1/credentials/{id}:revoke	FACILITATOR, ADMIN	Adds to revocation set; bumps registry version (epoch only on key compromise).
POST /v1/cohorts, GET /v1/cohorts/{id}/dashboard	FACILITATOR	Roster, missing documents, assessments, completions.
POST /v1/admin/rules/{id}:approve	REVIEWER	Adds a reviewer signature; pack becomes publishable at 2 of 3 and green golden tests.
POST /v1/credits/pools, /allowances, /settle	SPONSOR, HUB, DEVICE	Pool creation; lease issuance; settlement of receipts with dedup and cap checks.
GET /v1/audit	ADMIN	Hash-chained audit entries.

5.2.3 Data model

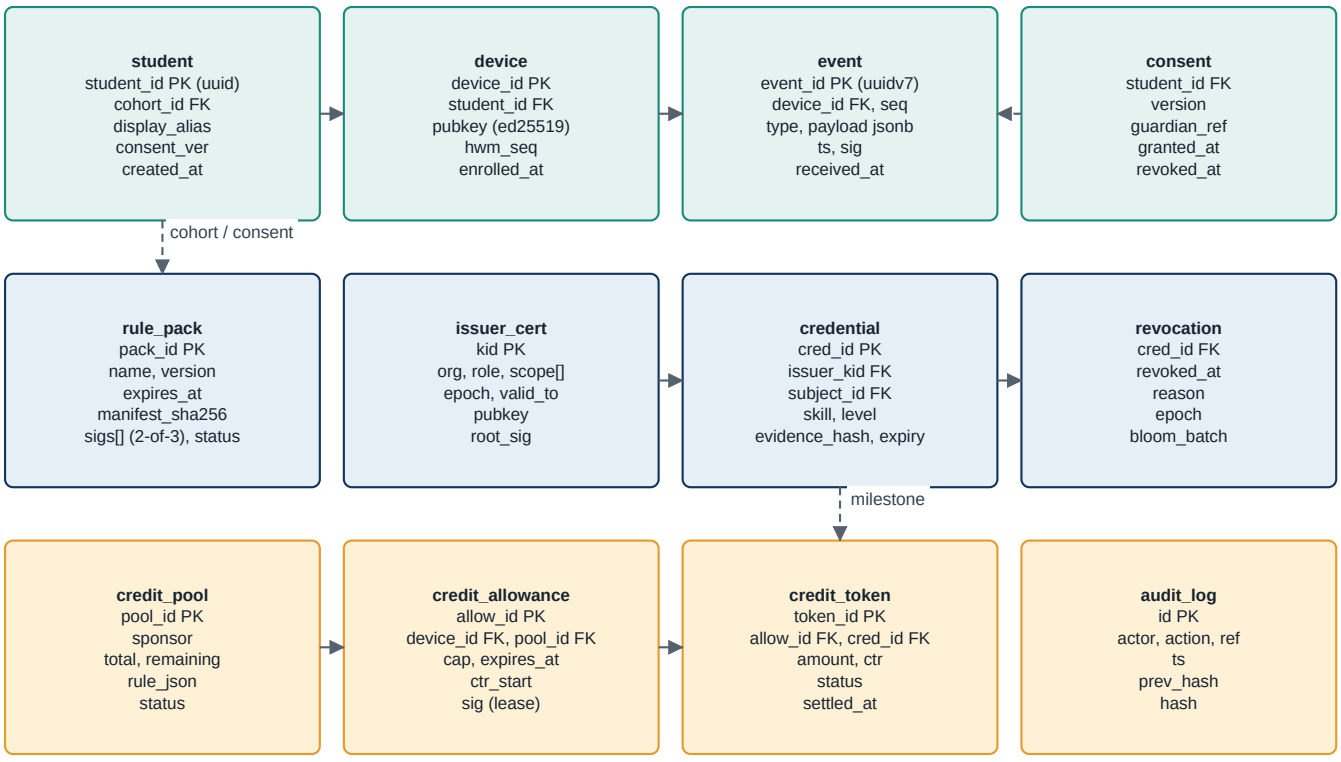


Figure 7. Logical data model (cloud). The hub uses the same schema in H2.

```
CREATE TABLE event (
  event_id    uuid PRIMARY KEY,                    -- UUIDv7, idempotency key
  device_id   uuid NOT NULL REFERENCES device,
  seq         bigint NOT NULL,                     -- per-device monotonic
  type        text   NOT NULL,
  payload     jsonb  NOT NULL,
  ts          timestamptz NOT NULL,
  sig         bytea  NOT NULL,                     -- device Ed25519 over canonical bytes
```

```
  received_at timestamptz NOT NULL DEFAULT now(),
  UNIQUE (device_id, seq)
);
CREATE TABLE credit_token (
  token_id uuid PRIMARY KEY, allow_id uuid NOT NULL, cred_id uuid NOT NULL,
  amount int NOT NULL CHECK (amount > 0), ctr bigint NOT NULL,
  status text NOT NULL CHECK (status IN ('MINTED','REDEEMED','SETTLED','FLAGGED')),
  UNIQUE (allow_id, ctr)
);
```

5.2.4 Security in the service layer

- JWT (short-lived) for sessions; every write also carries a device or issuer signature, so a stolen JWT cannot forge events.
- Server-side ownership checks on every cohort and student resource; facilitators see only their cohorts; sponsors see aggregates.
- Signing keys live outside the repo (environment secrets); root key is offline and used only to sign issuer certs and the pack key; rotation by epoch.
- Rate limiting and body-size limits on sync; replay protection by event_id uniqueness and per-device sequence.

5.3 FastAPI AI and tooling service

Endpoint	Behaviour	Guardrail
POST /v1/rules:extract	Takes scheme text or PDF; chunk; LLM emits structured rule JSON; validate against schema; every criterion must quote a source span found verbatim in the document	Output status is PROPOSED only; cannot reach a pack without reviewer signatures
POST /v1/profile:extract	Transcript to slots with confidence	Schema-validated; uncertain slots returned as UNKNOWN; student confirms
POST /v1/explain	Rule result to simple local-language text	Grounded on the result object only; template fallback; unsupported-claim check
POST /v1/packs:validate	Schema + golden-test run for a candidate pack	Deterministic; blocks publish on failure
CLI eval/fairness	Counterfactual profiles (change gender, region, language) and diff outcomes	Differences must map to an actual rule or be investigated

Bounded agents as a DAG with a human gate. The five Pagdandi agents are implemented as nodes in one DAG: (1) Profile and Conversation, (2) Career Discovery, (3) Pathway Planner, (4) Opportunity Matcher, (5) Explanation and Localisation. A confirm node (human in the loop) sits after node 1; node 4 calls the rules engine as a tool and cannot declare eligibility. The same DAG definition has two executors: a deterministic state machine on the device (templates, no LLM) and the FastAPI orchestrator online (LLM allowed for nodes 1 and 5). Retrieved documents are treated as untrusted data, tool arguments are schema-validated, calls have timeouts, and PII is stripped before any external call.

5.4 Wire formats

All artifacts use one **signed envelope**: the signature covers the exact payload bytes as sent, so no JSON canonicalisation is needed.

```
Envelope  = { "v":1, "kid":"<8-byte key id>", "typ":"pack|cred|registry|event|token|allowance",
              "payload":"<base64url bytes>", "sig":"<base64url Ed25519 over payload bytes>" }

RulePack payload (JSON):
{ "name":"schemes-mh", "version":"2026.10.14-1", "issued_at":"2026-10-14", "expires_at":"2026-11-13",
  "reviewers":["r1","r3"], "manifest":[{"file":"post-matric.json","sha256":"..."}, ...],
  "schemes":[{ "id":"PMS-OBC-MH", "name":{"en":"...","hi":"..."},
      "source":{"url":"https://...","verified":"2026-10-10"},
      "criteria":{"all":[ {"id":"c1","logic":{"<=":[{"var":"family_income"},250000]},"needs":["family_income"]},
                          {"id":"c2","logic":{"in":[{"var":"category"},["OBC","SC","ST"]]},"needs":["category"]} ]},
      "documents":["income_certificate","caste_certificate"], "deadline":"2026-12-31" }],
  "golden":[ {"profile":{...},"expect":"ELIGIBLE"}, {"profile":{...},"expect":"NOT:c1"}, {"profile":{},"expect":"NEED:[family_income,c

Credential payload:
{ "id":"<uuid>", "iss":"<issuer kid>", "epoch":3, "sub":"<student uuid>", "skill":"TAILOR.L2", "level":2,
  "ev":"<sha256 of evidence>", "iat":1760400000, "exp":1823472000 }          // about 350 bytes JSON + 64 byte sig

Credit token payload:
{ "tok":"<uuid>", "allow":"<uuid>", "cred":"<uuid>", "amt":150, "ctr":17, "iat":1760400000 }
```

QR budget. A credential envelope is well under 1,000 bytes and fits one QR at medium error correction. Larger artifacts (packs, event bundles) travel as files; animated multi-frame QR is a stretch option.

5.5 Algorithms

5.5.1 Three-valued eligibility

```
eval(node, profile):
  if node is leaf:
      for f in node.needs: if f not in profile (or confidence < threshold): collect f; return UNKNOWN
      return TRUE if jsonLogic(node.logic, profile) else FALSE
  if node.op == ALL:  r = [eval(c) for c in node.children]
      if any FALSE   -> FALSE (record first failing rule id)
      elif any UNKNOWN -> UNKNOWN (union of missing fields)  else TRUE
  if node.op == ANY:  TRUE if any TRUE; elif any UNKNOWN -> UNKNOWN; else FALSE
outcome: TRUE -> ELIGIBLE | FALSE -> NOT_ELIGIBLE(rule_id) | UNKNOWN -> NEED_INFO(fields, documents)
```

5.5.2 Offline credential verification

```
verify(envelope, registry, now):
  1. check typ == "cred"; decode payload
  2. cert = registry.issuer(payload.iss);  if none -> INVALID(unknown_issuer)
  3. ed25519.verify(cert.pubkey, envelope.payload, envelope.sig)  else INVALID(bad_signature)
  4. payload.epoch >= cert.epoch AND cert.valid_from <= payload.iat <= cert.valid_to  else INVALID(fenced_or_outside_lease)
  5. now <= payload.exp  else INVALID(expired)
  6. if registry.bloom.mightContain(payload.id): -> POSSIBLY_REVOKED (check exact list if cached, else "verify online")
  return VALID(registry.syncedAt)        // registry age always displayed
```

5.5.3 Idempotent sync push (server)

```
push(device, events[<=200]):
  verify JWT and device registration; rate limit
  rejected = []
  for e in events ordered by seq:
      if !schemaOk(e) or !verifySig(device.pubkey, e): rejected += (e.id, code); continue
      INSERT INTO event ... ON CONFLICT (event_id) DO NOTHING        -- duplicate is success
      apply projections (progress, profile fields via version vectors, credit receipts) in same transaction
  hwm = highest seq such that all seq <= hwm are stored or permanently rejected    -- contiguous high-water mark
  return { acked_upto: hwm, rejected, server_time }
```

5.5.4 Credit release and settlement

```
mint (hub):    require valid credential matching pool.rule.milestone; require allowance not expired;
               require ctr == last_ctr + 1 and spent + amt <= allowance.cap; sign token
redeem (dev):  verify token sig + allowance sig + not expired + ctr unseen on device; unlock module; store receipt
settle (cloud):require UNIQUE(allow_id, ctr); token_id not previously SETTLED on another device;
               sum(amounts) <= allowance.cap and pool.remaining >= amt
               violation -> status FLAGGED, loss bounded by cap; audit entry; sponsor sees aggregate only
```

5.5.5 Bloom filter for revocations (sizing)

For n revoked credentials and false-positive rate p, the filter needs $m = -n \ln p / (\ln 2)^2$ bits and $k = (m/n) \ln 2$ hashes. For n = 10,000 and p = 1%: m is about 95,850 bits (about 12 KB) and k is about 7. A negative answer is definitive (no false negatives), so devices can say "not revoked as of date". A positive answer is treated as "possibly revoked": the app shows a warning and checks the exact revoked list when cached, otherwise asks the verifier to confirm online. This keeps the registry delta small enough to move by file or QR set.

5.6 Key sequences and state machines

Sequence A: offline eligibility

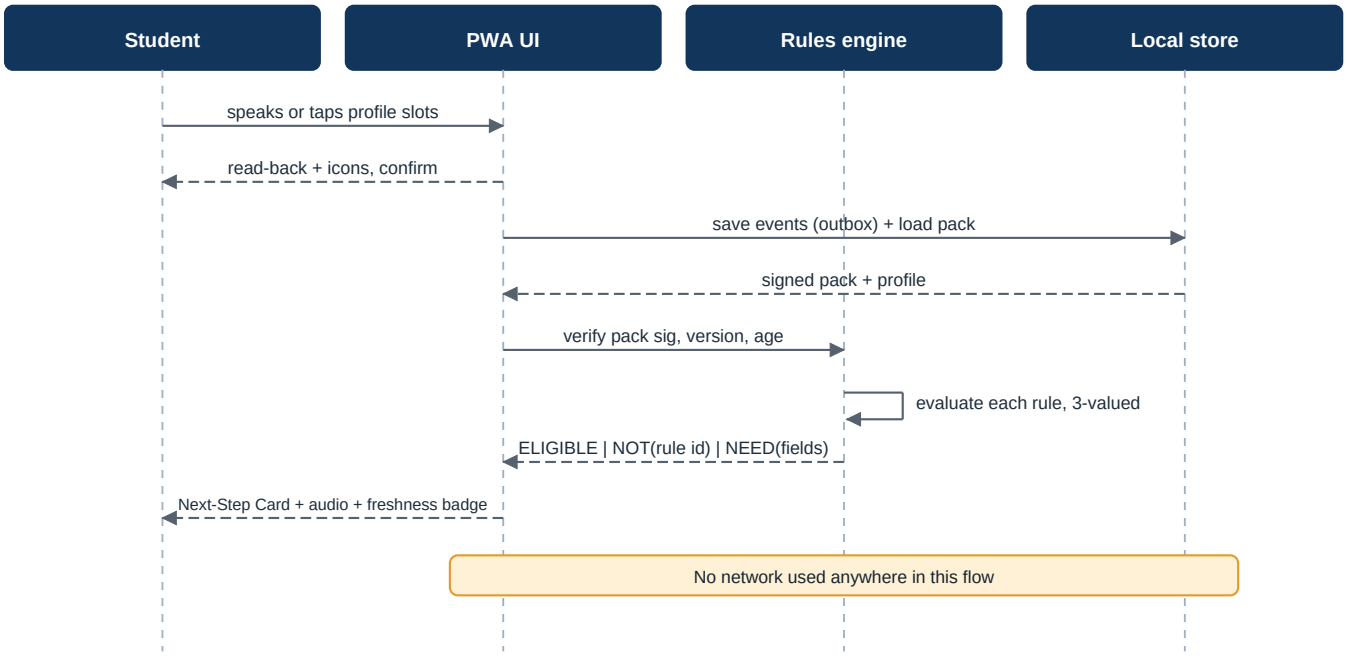


Figure 8. Voice profile to Next-Step Card without network.

Sequence B: credential issue and offline verification

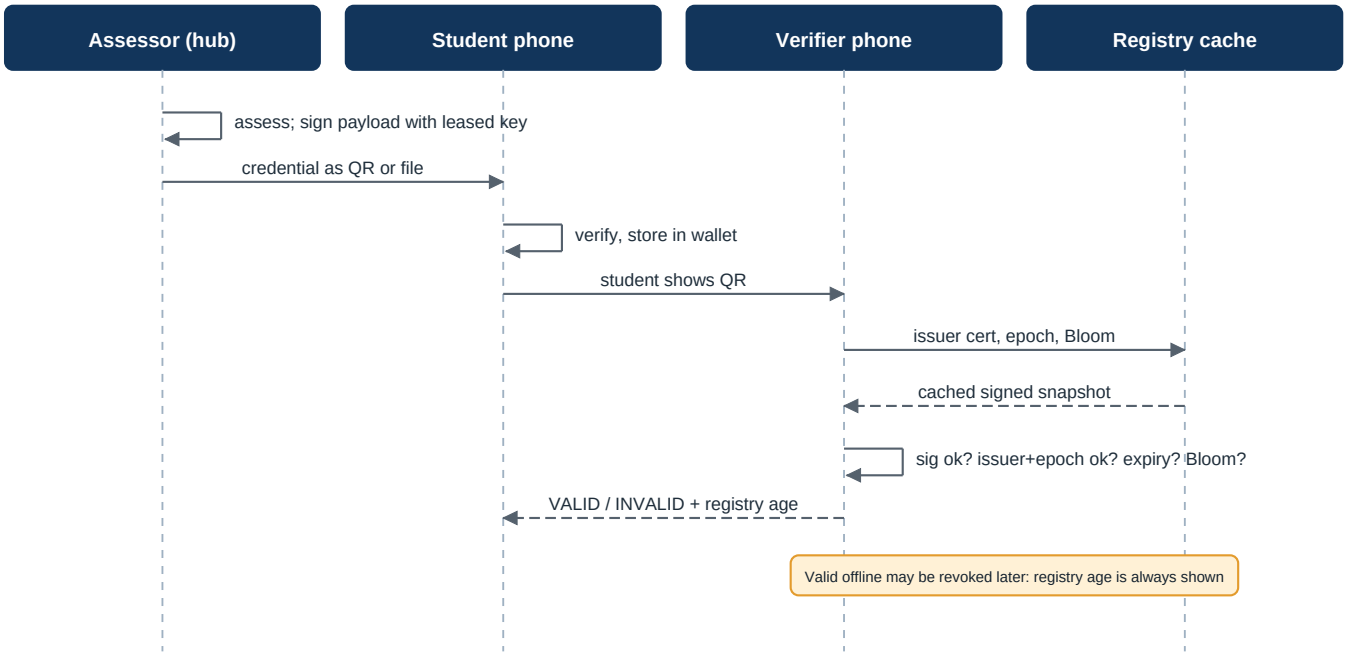


Figure 9. Issue under a leased key, verify against a cached registry.

Sequence C: idempotent sync and pack pull

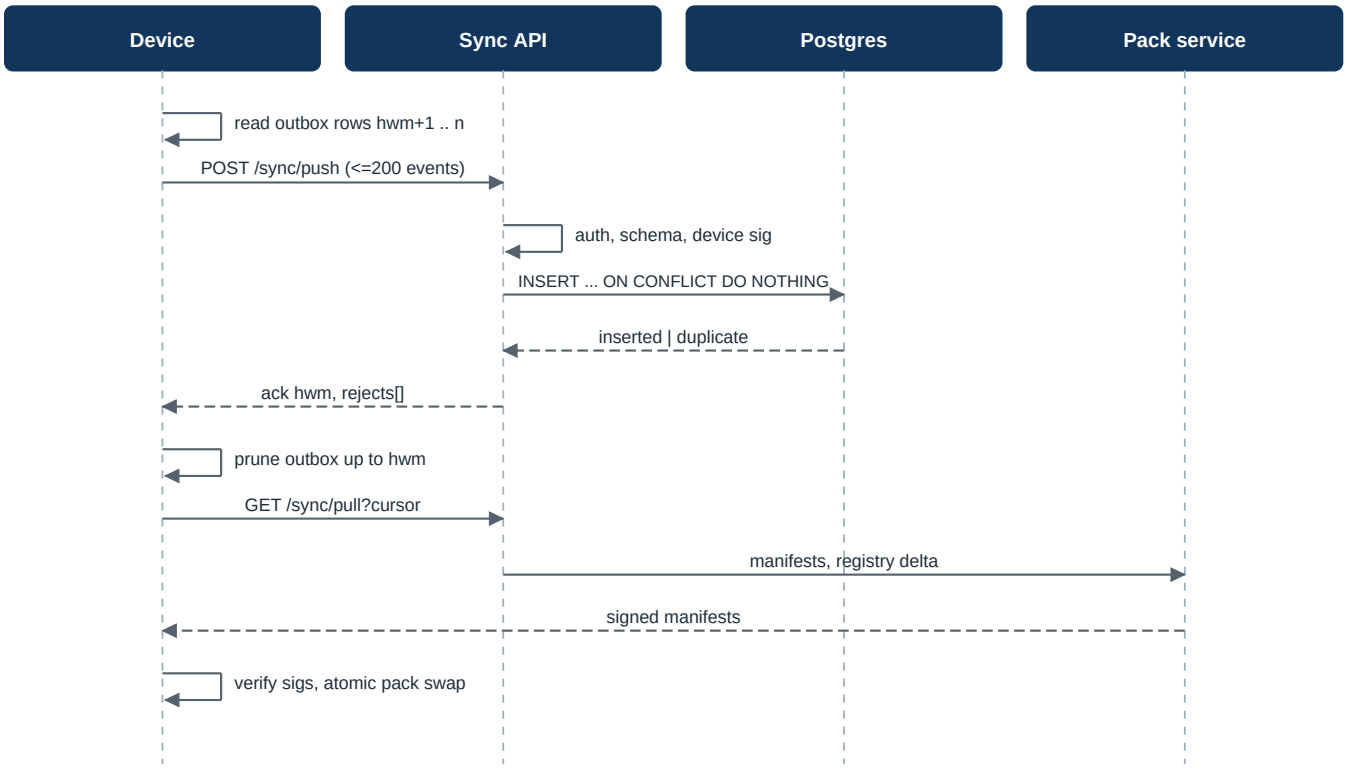


Figure 10. Push events with a high-water mark, then pull signed manifests.

Sequence D: pack update over an untrusted channel

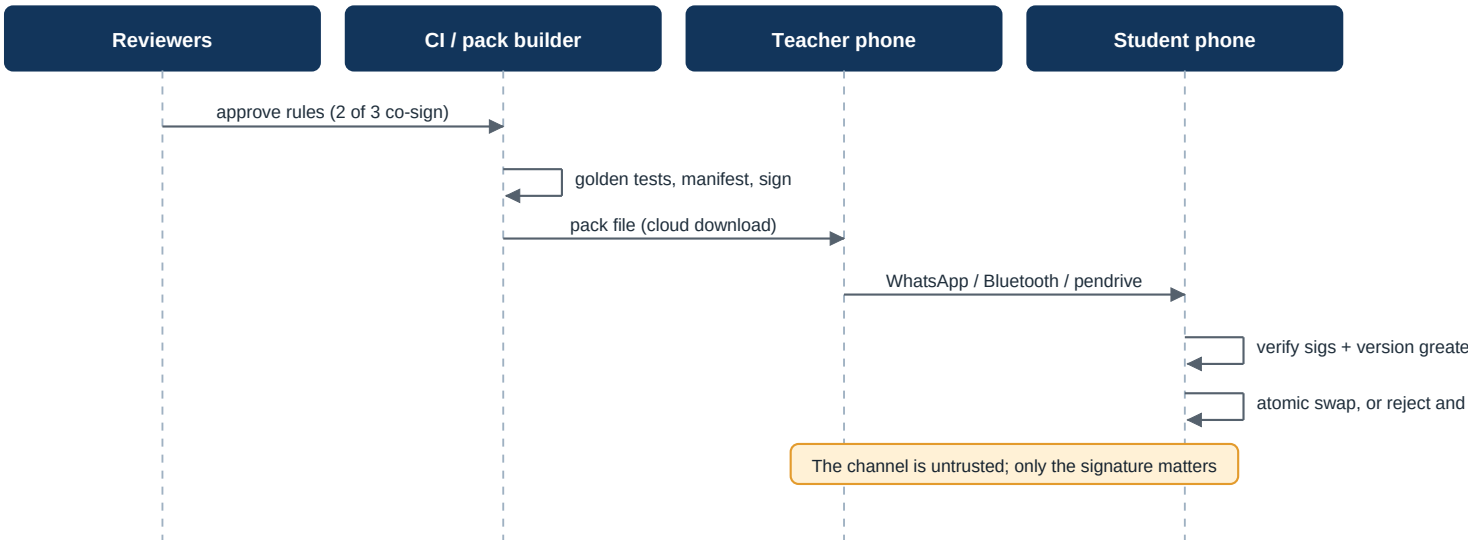


Figure 11. A signed pack forwarded by WhatsApp is accepted only if the signature and version check pass.

Sequence E: outcome-linked credit release

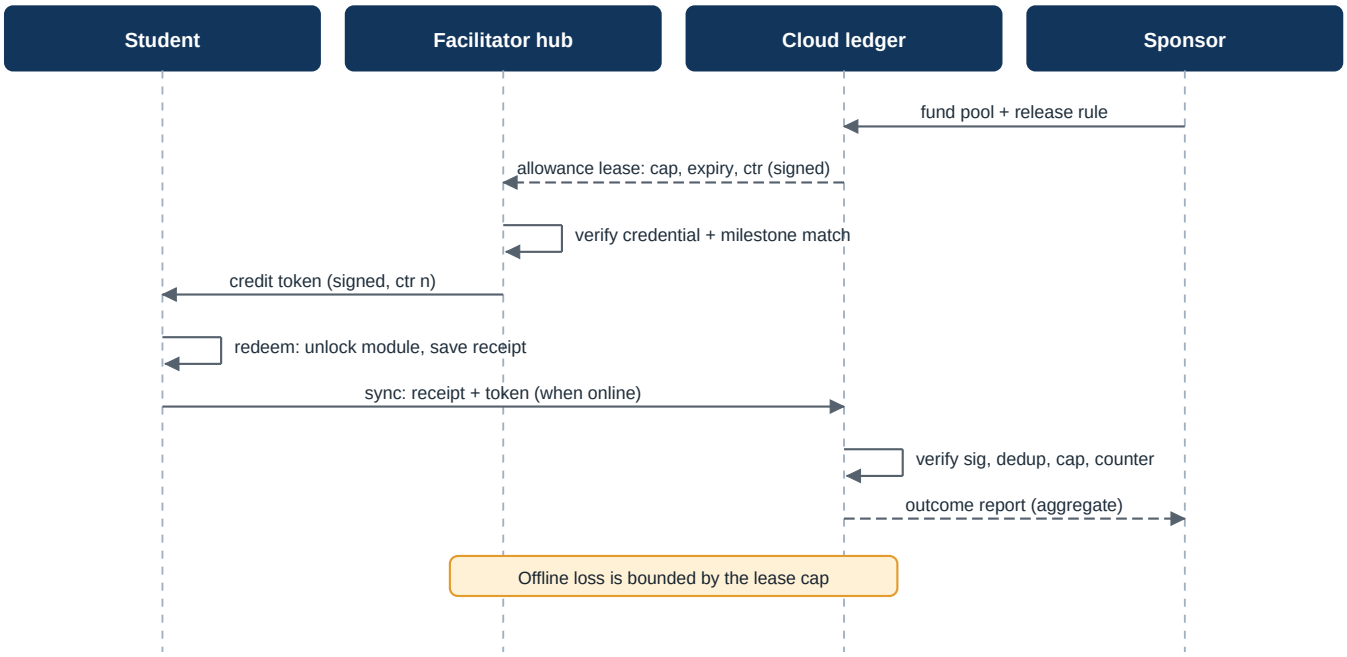


Figure 12. Sponsor funding to offline redemption and settlement.

State machines

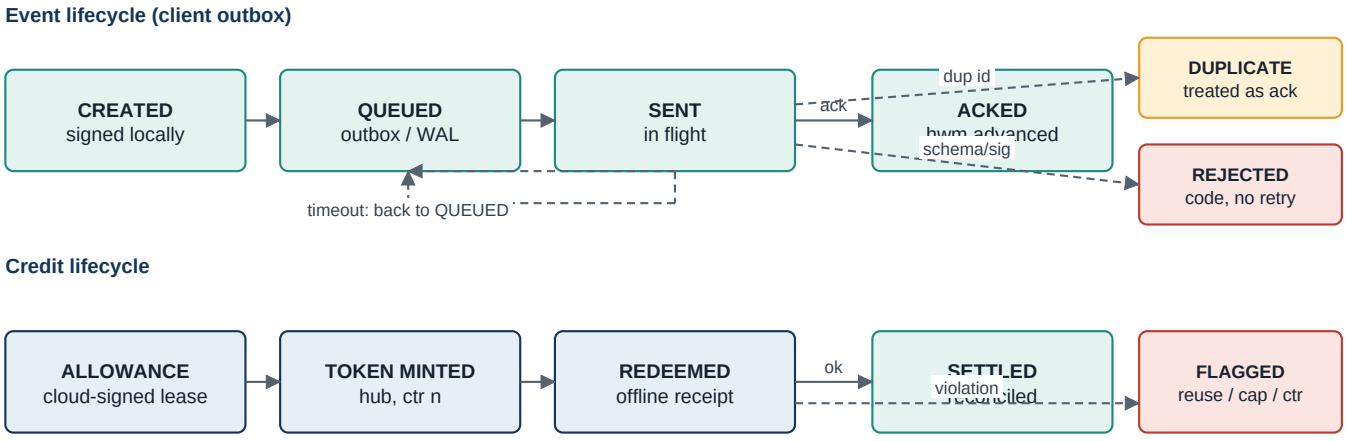


Figure 13. Event and credit lifecycles.

5.7 Hub profile and transports

Transport	Carries	Works offline	Status
Cloud HTTPS sync	Events up, packs/registry down	No (needs connection)	Must
QR code	Credentials, small tokens, short registry deltas	Yes	Must
Signed file via WhatsApp, Bluetooth, pendrive	Packs, event bundles (.sopaan), registry snapshots	Yes	Must (demonstrated in demo step 5)
Animated multi-frame QR	Medium artifacts without any radio	Yes	Stretch
LAN sync phone to hub	Events, cohort data	Yes, on a hotspot	Stretch; needs locally trusted HTTPS

Hub operation. The facilitator runs `docker compose up`, which starts Spring Boot (hub profile, H2 file database) and optionally FastAPI. The facilitator uses the hub UI at `http://localhost`, which is a secure context on that machine, so the same PWA and service worker work. The hub pulls packs and the registry when online, holds events until the cloud is reachable (hinted-handoff analogue), and issues credentials under a leased issuer cert that expires unless renewed at sync.

6. Failure, security and responsible-AI analysis

6.1 Failure modes and responses

Failure	What the user sees	System response	Test
No network	Everything in the offline column of 4.7 works; new actions show "pending sync"	Outbox queues; retry with backoff on reconnect	Playwright offline
Rule pack past expiry	Result shown with "Stale: pack from <date>" badge	Warn, never block; fetch new pack when possible	Clock-control UI test
Registry old	"Verified offline, registry as of <date>"	Show age; Bloom positive means possibly revoked	Verify unit tests
Tampered credential	INVALID: bad signature	Reject; log locally	Mutation tests on payload bytes
Bad or older pack received	"Update rejected, still using version X"	Atomic swap only after full verification; keep previous	Pack fuzz tests
Duplicate event after retry	Nothing	UNIQUE(event_id); duplicate treated as success	Replay test
Server cold start (Render)	Nothing (background sync)	Client timeout 90 s, exponential backoff	Latency injection
AI provider down	Template explanation; icon forms	Fallback chain: cloud ASR, offline STT, icons	Fault injection
Credit redeemed on two offline devices	Later flagged at settlement	Cap bounds loss; FLAGGED; audit entry	Double-spend test
Issuer key compromised	Old credentials show fenced/invalid after sync	Epoch bump rejects older-epoch signatures	Key-rotation drill
Two hubs issue with one key (split brain)	Counter conflict at sync	One lease per key; per-issuer counter; reject on collision	Concurrency test
Shared or lost device	PIN prompt; separate profiles	Wrapped device key; remote deauthorise device at next sync	Security review

6.2 Threat model (summary)

Threat	Attack	Mitigation
Spoofing	Forged credential or pack	Ed25519 signatures; issuer registry; key epoch fencing; pinned root key in app
Tampering	Altered QR or file in transit	Signature covers payload bytes; manifests hash each file
Repudiation	Issuer denies issuing	Server countersigned registry log; audit hash chain
Information disclosure	Student profile exposed to employers or sponsors	Consent-gated sharing; sponsor aggregates only; pseudonymous IDs; encrypt at rest and in transit
Denial of service	Sync flooding on a 0.1 CPU free server	Batch limits, rate limits, per-device quotas, offline-first so users are unaffected
Elevation of privilege	Facilitator sees other cohorts; AI writes rules	Server-side ownership checks; role guard; LLM output is PROPOSED only
Prompt injection	Malicious text in a scheme document	Documents treated as data; schema-validated output; citation spans verified; human review
Fraud	Fake training provider or opportunity	Verified issuers and providers; report button; no unverified listings shown as official

What a signature does not prove. A valid signature proves who signed and that nothing changed. It does not prove the assessment was fair or the issuer trustworthy. That is why issuers are authorised, scoped by skill, leased with expiry, and assessed against rubrics.

6.3 Privacy, child safety and fairness

- **Minimisation:** pseudonymous student IDs; the real-name roster stays with the facilitator; no Aadhaar by default; sponsor view is aggregate only.
- **Consent and minors:** versioned consent records and guardian consent for minors. Review the Digital Personal Data Protection Act 2023 provisions on children before any pilot.
- **Audio:** raw audio does not leave the device by default; cloud speech is opt-in per session.

- **Fairness tests:** counterfactual profiles (change gender, region, language) must produce equivalent recommendations unless an actual rule differs; report the pass rate and investigate every unexplained gap. A diversity rule keeps vocational and non-STEM options visible.
- **Transparency:** a "why this result" view with rule id, source and last-verified date on every opportunity; confidence indicators on extracted profile fields.
- **No harmful side effects:** no public rankings; talent indicators are visible to the student, correctable and never used to deny access; students never pay for essential guidance.
- **Distress path:** documented escalation; any helpline number is verified against current official sources before it is displayed.

7. Scalability and capacity

Scalability here means doing the heavy, frequent work on devices and scaling only what measurement proves is a bottleneck. The estimates below are back-of-envelope and must be replaced with load-test results.

7.1 Capacity estimate for a first pilot

Quantity	Assumption	Result
Pilot size	3 schools, about 1,000 students	1,000 pseudonymous students
Event volume	40 events per student per month	40,000 events per month, about 1,300 per day
Storage growth	400 B payload, about 3x for indexes and overhead	About 50 MB per month; Neon free (0.5 GB) lasts roughly 10 months before compaction or an upgrade
Sync burst	One class of 60 syncs after a power cut; 25 KB per phone	About 1.5 MB total; trivial. A whole school of 300 is about 7.5 MB over a few hundred batched calls
Credential QR	About 350 B JSON + 64 B signature	Fits one QR (< 1,000 B budget)
Revocation set	10,000 revoked, p = 1%	Bloom filter about 12 KB

7.2 Scaling triggers and responses

Measured symptom	Response
p95 API latency or error rate exceeds budget under burst	Scale stateless Spring workers horizontally; batch sync clients; admission control
AI cost or latency too high	Pre-translate and cache; pre-generated audio; route to cheaper or self-hosted models; template fallback
Event ingest or consumers need independence	Move from transactional outbox to a log such as Kafka
Repeated reads of scheme and registry data dominate	Add Redis cache, then read replicas; CDN for signed packs
One module blocks team velocity or scaling	Extract that module into its own service
Data outgrows one database	Shard ingest by district_id or hashed student_id (consistent hashing to place shards)
Pack distribution to many devices	Signed delta updates and CDN delivery

The target of 1,000 concurrent simulated users is a reproducible load test with a stated workload and error budget, not a claim about current production capacity.

7.3 Evolution path

Stage	Shape	Gate to move on
Prototype	PWA + Spring + FastAPI + Neon, Docker Compose, free tiers	Must flows pass offline and sync tests
Pilot (2 to 3 institutions, one district)	Managed containers, monitoring, backups, access-control testing, 3 validated languages	Signed agreements; usability and accuracy results
Multi-district	Horizontal workers, cache and queue where measured, formal rule-review and issuer governance, regulated payment partner if sponsors want real disbursement	Measured bottlenecks and compliance review

8. Quality: testing, metrics and observability

8.1 Test strategy

Layer	Technique	What it proves
Rules	Golden test suite shared by TypeScript and Java; property tests (fast-check, jqwik); three-valued edge cases	Client and server agree; missing data never becomes yes or no
Crypto	Known-answer vectors; mutation of every payload byte; wrong-key and old-epoch cases	Tampering is rejected
Sync	Replay, duplicate, reorder, drop and timeout injection	Effectively-once business effects
Offline	Playwright with the browser context set offline; real airplane-mode run on a low-end phone	Must flows work without network
API	Contract tests from OpenAPI; ArchUnit module boundaries	No cross-module leakage
Security	Authorisation matrix tests; OWASP ZAP baseline; secret scanning	Unauthorised cases denied
AI	Field-level extraction accuracy; explanation faithfulness against the result object; fairness counterfactuals	AI stays inside its boundary
Load	k6 burst of simulated syncs	Capacity and latency under the stated workload

8.2 Metrics and acceptance criteria (targets, not results)

Each reported metric must state sample size, data source, language, device, network condition and method. Perfect agreement on a small test set does not prove every encoded scheme is correct; rule correctness, source freshness and official decision outcomes are measured separately.

Metric	Target	Validation
Eligibility agreement	100% on a reviewed golden set (stating its size and coverage)	Independently reviewed expected outcomes
Explainability	100% of results include rule id, version, source, date	Schema test
Credential integrity	Valid accepted; altered or wrong-signer rejected	Cryptographic tests
Offline completeness	All Must flows in airplane mode	Device test
Sync	>= 99% valid events acked; zero duplicate effects	Fault injection
Credit release	Only after a valid credential; duplicates flagged	Ledger tests
Offline slot-STT accuracy	Word error rate under 15% on a defined closed-vocabulary test set (report separately from open dictation)	Hand-transcribed recordings
Recommendation quality	Recall@5 >= 0.80 and >= 3 non-engineering options when they exist	Human-labelled set
Explanation faithfulness	>= 0.90	Source-by-source review
Fairness	Report counterfactual pass rate; explain every gap	Counterfactual suite
Usability	>= 80% complete the primary journey; teacher time saved measured	Moderated tests with consent
Performance	Cold start < 5 s on 2 GB Android; verify < 200 ms; p95 warm API < 500 ms	Real devices, k6
Cost	< Rs 3 per student per month variable AI cost	Provider quotes and usage logs

8.3 Observability

- Structured JSON logs and Micrometer metrics from Spring Actuator; log-based dashboards on the free tier.
- Metrics: sync failures, event rejects by code, stale-pack and stale-registry counts, invalid-signature rate, rule errors, model latency, provider failures, credit FLAGGED rate.
- Client counters (opt-in, aggregate): offline completion, verify latency, STT fallback rate. No raw audio or free text.

9. Delivery: scope, plan, demo and risks

9.1 Scope tiers

Tier	Items
MUST (demo)	PWA with offline cache; 8 to 10 real, reviewed scheme rules with sources; shared golden-test parity (TypeScript and Java); signed packs; credential issue and QR verify; tamper and stale-registry cases; outbox with idempotent sync; closed-grammar voice in one language; Next-Step Card; facilitator dashboard; simulated (non-cash) credits; signed pack import by file.
STRETCH	FastAPI rule extraction with reviewer queue; 2-of-3 co-signing UI; fairness CI; animated-QR transfer; LAN hub sync with trusted HTTPS; Family Voice in a second language.
PLANNED (label as such)	Real payments or UPI payouts, regulated lender, DigiLocker/NSP/employer integrations, multi-language scale, Kafka/Redis/vector search, knowledge tracing.

9.2 Build order

Step	Deliverable	Done when
1	Shared rule schema, 3-valued evaluator in TypeScript and Java, golden suite in CI	Both pass identical vectors
2	Envelope, Ed25519 in three runtimes, pack signing and verification	Cross-language sign/verify tests green
3	Credential issue, registry snapshot, Bloom, QR encode/decode	Offline verify demo works on two phones
4	PWA shell, Dexie schema, WAL outbox, service worker	Airplane-mode e2e passes
5	Spring sync endpoints, idempotency, high-water mark	Replay test shows no duplicates
6	Voice grammar pipeline, icon fallback, audio clips	Profile completes by voice and by icons
7	Hub profile, cohort dashboard, simulated credits	Credit unlocks only after a valid credential
8	FastAPI rule extraction, fairness harness, polish, metrics panel	Reviewer approves a proposed rule end to end

9.3 Live demo: Radha's story

- **1. Discover.** Hindi voice profile; the app reads back what it understood with icons and she confirms.
- **2. Verified eligibility.** A scholarship result shows the deciding rule, source, version and missing documents.
- **3. Next step.** Next-Step Card and a short Family Voice audio.
- **4. Airplane mode.** The same check runs from the cached pack.
- **5. Pack by WhatsApp.** A newer signed pack arrives as a file while offline; the app verifies and switches.
- **6. Prove.** The facilitator issues a credential; a second phone verifies it from a QR.
- **7. Failure cases.** A tampered credential is rejected; a stale-registry warning appears.
- **8. Money follows proof.** A simulated sponsor credit unlocks only after the valid credential; reconnect, sync, replay an event, and nothing duplicates.

9.4 Risk register

Risk	Likelihood / impact	Mitigation
Scope too large for the time	High / High	Strict tiers; cut order: Family Voice 2nd language, LAN sync, FastAPI extraction last
JVM slow on Render free (0.1 CPU)	High / Medium	Lean Spring, -Xmx256m, background sync, demo on local hub
Offline Hindi STT accuracy unproven	Medium / High	Closed grammar, icon fallback, report WER honestly
Scheme rules wrong or stale	Medium / High	Source URL and date per rule, 2-person review, expiry on packs, report rule correctness separately
Free-tier terms change again	High / Medium	Re-verify in submission week; docker-compose fallback; portable stack
PWA install and secure-context limits	Medium / Medium	One online install or APK/TWA; document the limit
Key handling mistakes	Low / High	Keys in secrets, root key offline, epoch rotation drill
Evidence placeholders left empty	Medium / High	Replace every bracket with a verified source or interview before submission

10. Business model, Citi alignment and judging map

10.1 Revenue model (illustrative planning assumptions to validate in a pilot)

Payer	Pays for	Illustrative
Students and families	Nothing; essential guidance and Skill Passport are free	Free
NGOs, CSR programmes, education partners	Cohort tools, verified outcomes, aggregate impact reporting	About Rs 60 per student per year
Employers and apprenticeship providers	Consent-based verified candidates and credential verification	About Rs 500 to 1,000 per enrolled candidate
Training providers and institutions	Listings, issuer onboarding	To validate
Facilitators (earn)	Module sales and commissions	Keep the majority; platform share 10 to 15%

Unit economics example: at 8 voice turns per student per month and an assumed Rs 0.80 per cloud turn, live APIs would cost about Rs 6.40 per student per month; because offline-first and pre-generated audio avoid most cloud turns (assume 60% avoided), variable AI cost is about Rs 2 to 2.60. These are planning numbers, not a profitability claim, and exclude engineering, support and taxes.

Fintech mechanism and limits. Outcome-linked learning credits release sponsor funds only against a verified milestone. In the prototype they are simulated, non-cash entitlements. Sopaan is not a lender, a scholarship authority or a payment provider, and a credential is one piece of evidence that does not approve a loan or act as a credit score. Any Citi, government, school or employer relationship is prospective until confirmed.

10.2 Mapping to the judging criteria

Criterion (weight)	How Sopaan answers it	Evidence in this document
Innovation and originality (25%)	One verifiable chain from rule to result to credential to sponsor release; signature-over-any-channel; AI proposes, rules decide; career widening by design	1, 3, 4.3
Technical feasibility and architecture (20%)	Modular monolith, deterministic engine, signed packs, offline-first PWA, idempotent sync, tested failure modes, free-tier deployment	4, 5, 6
Usability and scalability (20%)	Voice and icons, small bundles, heavy work on device, measured scaling triggers	5.1, 7
Monetization (15%)	Free for students; NGO/CSR, employer, provider, facilitator revenue; stated unit-cost target	10.1
Customer impact and UX (20%)	Radha goes from "I do not know what is possible" to a verified, provable next step	1, 9.3

Appendices

A. Glossary

Term	Meaning
Pack	A signed, versioned bundle of data (rules, catalogue, audio index, registry) that devices verify before use
Envelope	The common signed container: key id, type, payload bytes, Ed25519 signature
Issuer lease	A time-limited, scoped authorisation for a facilitator key to sign credentials; carries an epoch
Epoch / fencing token	Counter that invalidates older signatures from a compromised or superseded key
High-water mark	Largest contiguous event sequence the server has stored; the client prunes up to it
Outbox / WAL	Local queue of events written in the same transaction as the state change
Allowance	Cloud-signed lease giving a device a capped, expiring right to redeem credits offline
Next-Step Card	One screen per opportunity: status, deciding rule, missing documents, deadline, place, helper
Family Voice	Short local-language audio summary for the student's family, with consent
Hub	The facilitator laptop running the same backend in the "hub" profile

B. Pre-submission checklist

- Replace every evidence placeholder with a verified source (for example a UDISE+ statistic with year) or interview findings gathered with consent.
- Choose real schemes, verify official criteria and dates, and record the source URL and last-verified date in each rule.
- Pick one language and test the demo in real airplane mode on a low-end phone.
- Show one tampered credential, one stale-registry warning, one safe sync retry, one signed pack imported by file.
- Re-check free-tier limits, voice-provider credits and licences, and competitor claims in the submission week.
- Check the name Sopaan against trademark, domains and app stores.
- Describe Citi, government, school and employer relationships as prospective until confirmed.
- Keep targets and measured results separate on every slide and in every spoken claim.

C. Verification notes

Free-tier and pricing facts in section 4.6 were checked on 14 October 2026 from provider documentation and third-party summaries: Render (free web service limits and spin-down), Neon (free plan quotas), Oracle Cloud (Arm allowance reduced in 2026), Sarvam (speech pricing and free credits, with conflicting figures across pages), and Hugging Face Spaces documentation. These change frequently; treat them as a snapshot. Technical claims about Vosk grammar support, Hindi model size, and Android offline voices are design assumptions to confirm on the chosen builds before committing.